

A High-Power DC-DC Converter for Electric Vehicle Battery Charger

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Abstract— A DC-DC converter for high-power battery chargers with the power-capacity over 6.6-kW for electric vehicle (EV) is proposed in this paper. Due to a new architecture, the proposed converter achieves the reduction of conduction loss at the primary-side by as much as 50% and has many benefits such as reduced circulating current, less duty-cycle loss, and lower secondary-voltage stress. In addition, its power handing capacity can be upsized easily with the usage of two full-bridge inverters and two transformers. Besides, all the active switches in the proposed converter achieve zero-voltage-switching (ZVS) during the whole battery charging process, and the size of output filter can be significantly reduced. The circuit configuration, operation, and relevant analysis are presented, followed by the experiment on a prototype realized with a scale-downed 3.3-kW charger. The experimental results validate the theoretical analysis and show the effectiveness of the proposed converter as a battery charger.

Keywords— Full-bridge converter, electric vehicle, hybrid electric vehicle, battery charger, zero-voltage switching.

I. INTRODUCTION

Due to global warming and decreasing fossil fuel, electric vehicles (EVs) are growing quickly in the vehicle market. EVs need necessarily rechargeable energy storage such as batteries [1]. To extend the driving range, today, EVs require the battery with much higher capacity and the power-capacity of the charger for the battery is also continuously increased [2], [3].

The battery charger is composed of a power-factor corrector (PFC) and a DC-DC converter [4]–[8]. In the development of battery charger, it is necessary to decrease the size and mass. And, the power-conversion efficiency will be higher during the overall battery charging time. In order to acquire these, the usage of higher switching-frequencies is recommended with soft-switching technologies. In detail, a bridgeless design should be carried out at the PFC part, because an excessive loss is generated due to the large forward-voltage-drop at each of bridge diodes [9]–[11]. In order to achieve higher efficiencies, interleaved or parallel operation can be adopted since they decrease significantly the conduction-loss [12]–[16]. Besides, in the DC-DC converter part, reducing the voltage stress of diodes in the rectifier can contribute to the improvement of the conduction-loss [17].

In this paper, a DC-DC converter for high-power battery chargers with the power-capacity over 6.6-kW for EVs is

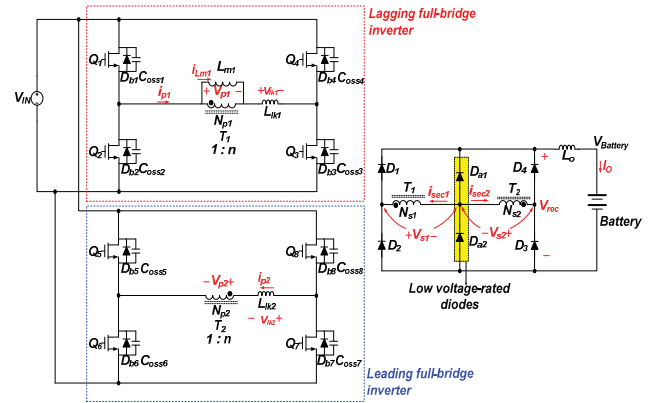


Fig. 1 Proposed DC-DC converter for high-power electric vehicle battery charger.

proposed. The proposed converter consists of two full-bridge inverters (TFBIs) in parallel at the primary-side. They are worked in phase-shift manner to control the output. Two full-bridge diode rectifiers are adopted for the rectifier, sharing a diode-leg with lower-voltage rating. Due to this architecture, the proposed converter achieves the reduction of conduction loss at the primary-side by as much as 50% and has many benefits such as reduced circulating current, less duty-cycle loss, and lower secondary-voltage stress. In addition, its power handing capacity can be upsized easily with the usage of two full-bridge inverters and two transformers. Besides, all the active switches in the proposed converter achieve zero-voltage-switching (ZVS) during the whole battery charging process, and the size of output filter can be significantly reduced.

II. OPERATION PRINCIPLE

The circuit configuration of the proposed converter is shown in Fig. 1 and Fig. 2 shows its key operating-waveforms. Two full-bridge inverters are worked in the phase-shift and the active switches in each full-bridge inverter are triggered with constant duty-ratio of about 50%. The gate signals for switches in the leading full-bridge inverter lead those for switches in the lagging full-bridge inverter. From now, switches in the leading

or lagging full-bridge inverters are called as leading or lagging switches, respectively.

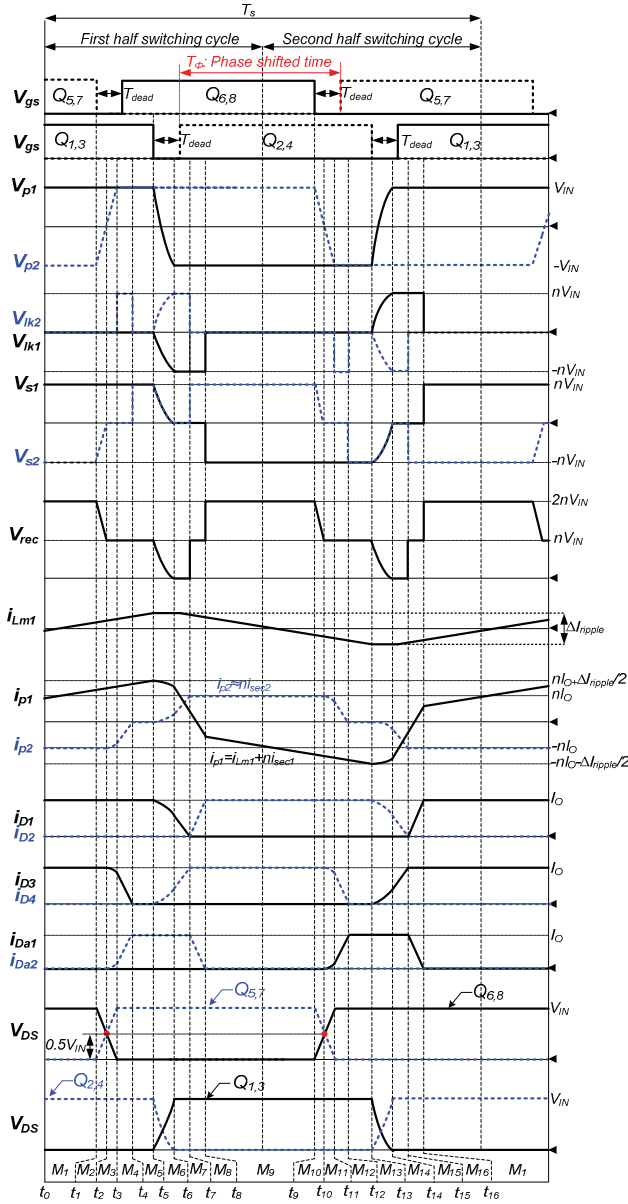


Fig. 2 Key operating waveforms of the proposed converter in steady state.

Each switching period is divided into two half cycles, t_0 - t_8 and t_8 - t_{16} . Because the operational principles of two half cycles are symmetric, only the first half cycle is described. This half cycle can be subdivided into eight modes, whose operating circuits are shown in Fig. 3.

Mode 1 [t_0 - t_1]: Mode 1 begins when $Q_{5,7}$ of leading switches and $Q_{1,3}$ of lagging switches are in ON-state and diodes D_1 and D_3 are conducting. In this mode, the voltages at the primary-side of transformers, $V_{p1}(t)$ and $V_{p2}(t)$, become positive and negative values of V_{IN} , respectively. And the magnetizing current of T_1 , $i_{Lm1}(t)$ starts linearly to increasing, but the $i_{Lm2}(t)$ is nearly zero due to the very large magnetizing inductance of T_2 . The voltages at the secondary-side of transformers, $V_{s1}(t)$ and $V_{s2}(t)$, also become positive and negative values of the input voltage with the turns-ratio of n , respectively. The output voltage of rectifier, $V_{rec}(t)$ becomes $2nV_{IN}$ due to the sum of $V_{s1}(t)$ and $-V_{s2}(t)$. In this mode, the power is transferred from

the input port to the output port via T_1 , T_2 , D_1 , and D_3 and the primary currents of inverter stage can be expressed as follows:

$$\begin{aligned} i_{p1}(t) &= i_{Lm1}(t) + n i_{sec1}(t) = i_{Lm1}(t) + n i_{D1}(t) = i_{Lm1}(t) + n I_O, \\ i_{p2}(t) &= i_{Lm2}(t) + n i_{sec2}(t) \approx n i_{sec2}(t) = -n i_{D3}(t) = -n I_O. \end{aligned} \quad (1)$$

Mode 2 [t_1 - t_2]: Mode 2 begins when $Q_{5,7}$ are turned off at t_1 .

Then, by the energy stored in the output inductor L_O , the voltages across C_{OSS5} and C_{OSS7} are charged linearly and the voltages across C_{OSS6} and C_{OSS8} are discharged. $V_{p2}(t)$ increases from $-V_{IN}$ and $V_{p1}(t)$ is maintained at V_{IN} . The $V_{rec}(t)$ falls from $2nV_{IN}$ to nV_{IN} . The expression of primary currents in this mode is the same as in mode 1.

Mode 3 [t_2 - t_3]: When the $V_{p2}(t)$ becomes zero, mode 3 starts. Then the $V_{s2}(t)$ is zero and the diode of D_{a2} begins to be ON. Because D_{a2} is ON, $V_{s2}(t)$ is maintained at zero in this mode and a resonance behavior between the output capacitances of all the leading switches and L_{lk2} occurs in the leading full-bridge inverter. By the resonance, the voltages across $C_{OSS5\&7}$ are continuously charged and the voltages across $C_{OSS6\&8}$ are discharged. The $V_{p2}(t)$ increases from zero to V_{IN} , and $V_{p1}(t)$ is maintained at V_{IN} .

Mode 4 [t_3 - t_4]: The $V_{p2}(t)$ becomes V_{IN} in mode 3 and then mode 4 starts. The body diodes of $Q_{6,8}$, D_{b6} and D_{b8} , start to conduct and $Q_{6,8}$ are turned on with ZVS. In mode 4, $V_{s2}(t)$ is continuously zero, thus the input voltage of V_{IN} applies to the leakage inductor of L_{lk2} . Due to the positive voltage applied to L_{lk2} , D_{a2} begins to commutate with D_3 . $V_{p1}(t)$ and $V_{rec}(t)$ are maintained at V_{IN} and nV_{IN} , respectively. In this mode, the input power is transferred to the output load through T_1 , T_2 , D_1 , D_{a2} , and D_3 , and the currents can be analyzed as follows.

$$\begin{aligned} i_{p1}(t) &= i_{p1}(t_3) + \frac{V_{IN}}{L_{m1}}(t-t_3), \quad i_{p2}(t) = -n I_O + \frac{V_{IN}}{L_{lk2}}(t-t_3), \\ i_{D1}(t) &= I_O, \quad i_{D3}(t) = I_O - i_{Da2}(t), \quad i_{Da2}(t) = \frac{V_{IN}}{n L_{lk2}}(t-t_3). \end{aligned} \quad (2)$$

Mode 5 [t_4 - t_5]: Mode 5 begins when the commutation between D_3 and D_{a2} is completed at t_4 and only D_1 and D_{a2} are conducting. In this mode, the $i_{p2}(t)$ is zero, and the input power is transferred to the output load through T_1 , D_1 , and D_{a2} .

Mode 6 [t_5 - t_6]: Mode 6 starts when $Q_{1,3}$ are turned off at t_5 . At the same time, diode D_4 starts to conduct and another resonance, which consists of the output capacitances of all the lagging switches and the two leakage inductors of L_{lk1} and L_{lk2} , occurs in the lagging full-bridge inverter. By the resonance, the voltages across $C_{OSS1\&3}$ are discharged, and the voltages across of $C_{OSS2\&4}$ are charged. The $V_{p1}(t)$ decreases from V_{IN} to $-V_{IN}$ and $V_{rec}(t)$ decreases to zero. The commutation between D_1 and D_4 is also progressed.

Mode 7 [t_6 - t_7]: When the $V_{p1}(t)$ becomes $-V_{IN}$ in mode 6, mode 7 begins. Then the body diodes of $Q_{2,4}$, D_{b2} and D_{b4} , start to be ON and $Q_{2,4}$ is turned on with ZVS. In mode 7, the $V_{s1}(t)$ and $V_{s2}(t)$ are zero, and thus $V_{rec}(t)$ is also zero. Due to this, the output load is powered from the energy stored in the output inductor L_O at t_6 . Because $V_{p1}(t) = -V_{IN}$, $V_{p2}(t) = V_{IN}$, and $V_{s1}(t) = V_{s2}(t) = 0$, the voltages across two leakage inductors, L_{lk1} and L_{lk2} , equal to $-V_{IN}$ and V_{IN} , respectively. Then the $i_{p1}(t)$ and $i_{D1}(t)$ decrease linearly and the $i_{p2}(t)$ and $i_{D4}(t)$ increase linearly

as follows.

$$i_{p1}(t) = i_{p1}(t_6) - \frac{V_{IN}}{L_{lk1}}(t - t_6), \quad i_{p2}(t) = i_{p2}(t_6) + \frac{V_{IN}}{L_{lk2}}(t - t_6),$$

$$i_{D4}(t) = I_O - i_{D1}(t) = i_{p2}(t)/n, \quad i_{D42}(t) = I_O. \quad (3)$$

Mode 8 [t_7 - t_8]: Mode 8 begins when the current through D_4 reaches to the output current I_O and D_1 is turned off. Then the $V_{s1}(t)$ is zero and $V_{s2}(t)$ becomes nV_{IN} . Thus, the V_{IN} appears on L_{lk1} and the commutation between D_2 and D_{a2} starts. The $V_{rec}(t)$ equals to nV_{IN} . The currents in this mode can be represented as follows.

$$i_{p1}(t) = i_{p1}(t_7) - \frac{V_{IN}}{L_{lk1}}(t - t_7), \quad i_{p2}(t) = nI_O, \quad i_{D4}(t) = I_O,$$

$$i_{D2}(t) = I_O - i_{D42}(t) = \frac{V_{IN}}{nL_{lk1}}(t - t_7). \quad (4)$$

At the end of mode 8, $i_{D2}(t)$ reaches to I_O and the diode of D_{a2} is turned off naturally. Then, the input power is transferred to the output load through the path of T_1 , T_2 , D_2 , and D_4 .

III. ANALYSIS RESULTS

A. Voltage gain

Since the intervals of mode 2, 6, and 8 in Fig. 2 are practically very narrow, the output voltage of the rectifier in the proposed converter, $V_{rec}(t)$ can be simply given as shown in Fig. 4(a). Fig. 4(b) shows the counter part for the conventional converter, which is single phase-shift full-bridge (PSFB) converter or a topology consisting of two parallel-operated PSFB converters for higher-power designs. Comparing the two figures, we can know that the proposed converter do not have the freewheeling time interval, where $V_{rec}(t)$ is maintained at zero. Due to this, the proposed converter transfers continuously the input power to the output load, and hence any circulating current is not generated in the proposed converter.

By averaging the $V_{rec}(t)$ in Fig. 4(a), the voltage gain for the proposed converter can be acquired as follows.

$$M(D) = \frac{V_O}{V_{IN}} = 2n(D + 0.5), \quad D = (T_\phi - 2T_{dead}) / T_S, \quad (5)$$

where T_S is a switching period.

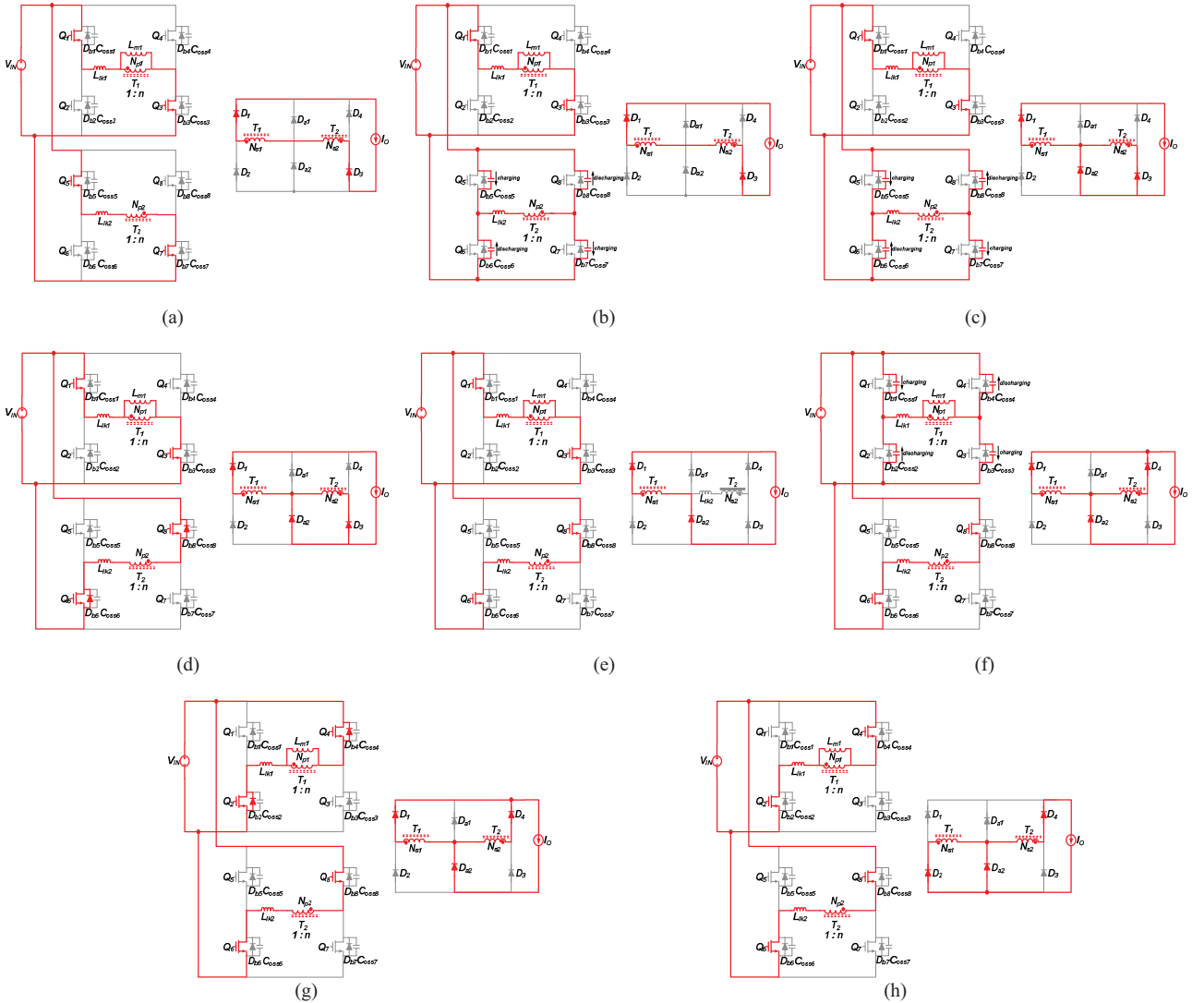


Fig. 3 Operating circuits during the first half cycle: (a) Mode 1, (b) Mode 2, (c) Mode 3, (d) Mode 4, (e) Mode 5, (f) Mode 6, (g) Mode 7, and (h) Mode 8.

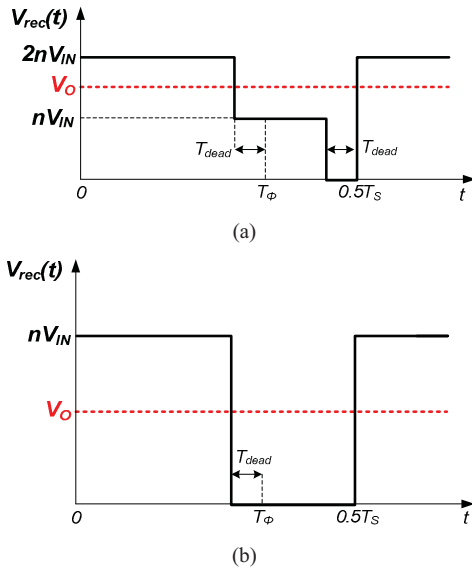


Fig. 4 Simplified rectifier output waveform (a) in the proposed converter and (b) in the conventional converter consisting of two parallel-operated PSFB converters.

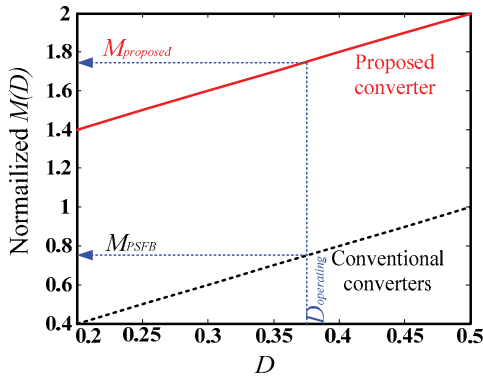


Fig. 5 Normalized DC conversion ratio versus duty-cycle.

From Fig. 4(b), the voltage gain for the conventional converter is obtained as

$$M(D) = \frac{V_o}{V_{IN}} = 2nD, \quad D = (T_\phi - T_{dead}) / T_s. \quad (6)$$

The normalized $M(D)$ (supposing $n=1$) of the proposed converter is shown in Fig. 5 along with the conventional converter. Fig. 5 notes that the proposed converter has much higher voltage gain than the conventional converter. Due to this, the proposed converter can be designed with much smaller turns-ratio n than the conventional converter. This contributes to the improvement of the conduction loss in the primary-side and the voltage stress in the secondary-side.

B. ZVS Condition

As described in the operating modes, the ZVS for the leading switches is formed through two resonant phases. Firstly, by the large energy stored in the output inductor, the drain-to-source voltage of the leading switches is discharged from V_{IN} to $0.5V_{IN}$ linearly. After that, the voltage are fully discharged by the resonance formed by the energy stored in the leakage inductor L_{lk2} . The ZVS condition for the leading switches can be expressed as follows.

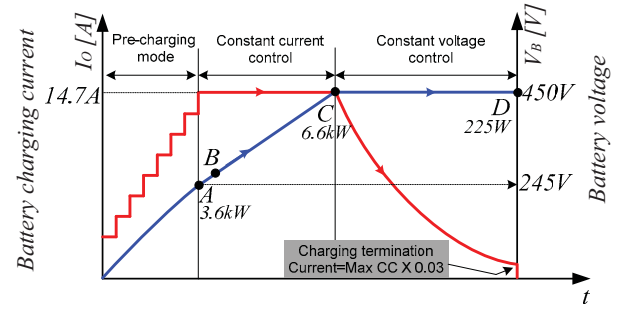


Fig. 6 Battery charging profile for 6.6-kW battery charger.

$$nI_o z_{O1} \sin \omega_{O1} (T_{dead} - \frac{V_{IN} C_{OSS}}{nI_o}) \geq 0.5V_{IN}. \quad (7)$$

In the view of energy, eq. (7) can be represented as

$$\frac{1}{2} L_{lk2} (nI_o)^2 \geq \frac{8}{3} C_{OSS} (0.5V_{IN})^2 = (\frac{8}{3} C_{OSS} V_{IN}^2) \times \frac{1}{4}. \quad (8)$$

As seen in (8), the voltage remaining at the leading switches, which should be removed by the energy stored in L_{lk2} , is only 25% of the original energy. Due to this, the ZVS operation for the leading switches can be acquired easily with the small inductance of L_{lk2} under wide load variations. If the current through L_{m2} is considered to ensure the ZVS, the ZVS equation in eq. (8) can be modified as in (9).

$$z_{O1} (0.5\Delta I'_{ripple} + nI_o) \sin \omega_{O1} (T_{dead} - \frac{V_{IN} C_{OSS}}{0.5\Delta I'_{ripple} + nI_o}) \geq 0.5V_{IN} \quad \text{or} \quad (9)$$

$$\frac{1}{2} L_{lk2} (nI_o + 0.5\Delta I'_{ripple})^2 \geq \frac{8}{3} C_{OSS} (0.5V_{IN})^2 = (\frac{8}{3} C_{OSS} V_{IN}^2) \times \frac{1}{4},$$

where $\Delta I'_{ripple}$ is the current ripple of L_{m2} . Because the energy that is discharged by $\Delta I'_{ripple}$ is also 25% of the original energy as shown in (9), $\Delta I'_{ripple}$ can be designed with a small value.

The ZVS condition of the lagging switches is obtained as follows.

$$z_{O2} (0.5\Delta I'_{ripple} + nI_o) \sin \omega_{O2} T_{dead} \geq V_{IN}. \quad (10)$$

In the view of energy, eq. (10) can be expressed as

$$\frac{1}{2} (L_{lk1} + L_{lk2}) (0.5\Delta I'_{ripple} + nI_o)^2 \geq \frac{8}{3} C_{OSS} V_{IN}^2. \quad (11)$$

From eq. (10) or eq. (11), it is known that the ZVS of the lagging switches can be easily achieved under wide load variations due to the term of $L_{lk1} + L_{lk2}$ and the ripple of $i_{Lm1}(t)$, $\Delta I'_{ripple}$. Typically, large leakage inductors increase the duty-cycle loss and hence a trade-off between the leakage inductor's values and magnitude of $\Delta I'_{ripple}$ should be conducted in (11) for optimally designing the proposed converter. In order to extend the ZVS range to no load condition with reasonable leakage inductors' values, the proposed converter uses the magnetizing inductor of T_1 , L_{m1} (or $\Delta I'_{ripple}$) in this paper. Then, someone could think that the conduction loss generated in the lagging full-bridge inverter will increase as in the conventional phase-shift full-bridge converter with a small magnetizing inductance. However, the RMS current in the lagging full-bridge inverter is not increased in spite of the small magnetizing inductance of T_1 .

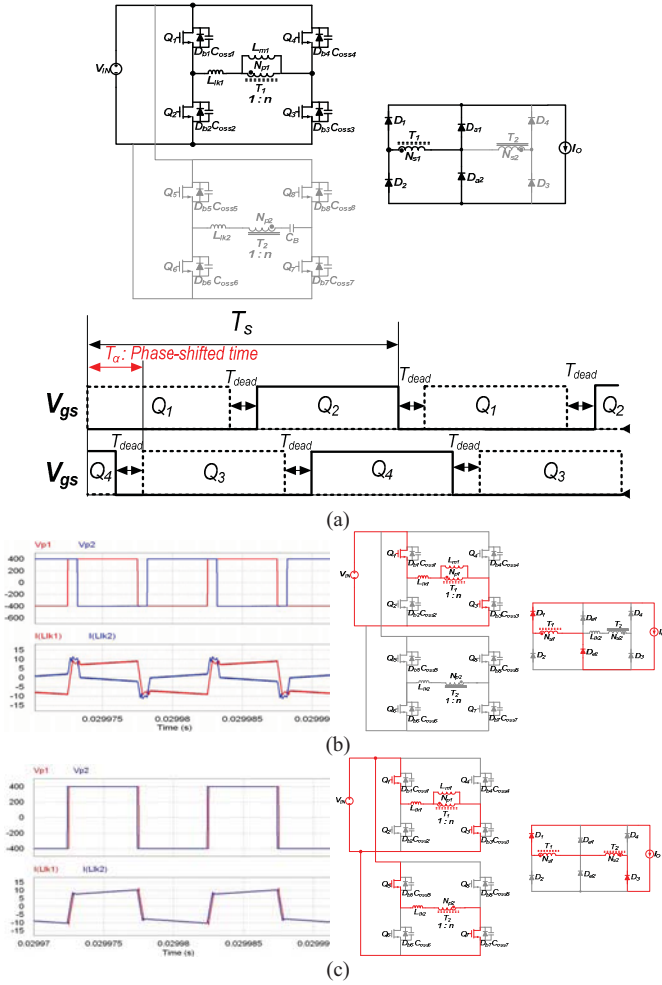


Fig. 7 Operations of the proposed converter over the battery charging profile in Fig. 7; (a) In the pre-charging mode, (b) At the point B, and (c) At the point C.

The average value of the magnetizing current in T_1 becomes zero within a half-switching period and thus its contribution to the RMS current is negligible at heavy load conditions. This principle is well explained analytically in [18].

C. Operation of the proposed converter over battery charging profile

Fig. 6 shows the battery charging profile for a 6.6-kW battery charger [19]. At the beginning, a DC-DC converter gradually ramps up charging current with a staircase wave to avoid

TABLE I
COMPONENTS LIST

Main switches (Q_1 – Q_6)	IPW65R080CFDA(650V)	
Rectifier diodes (D_1 – D_4)	IDH15S120(1200V, 15A, $V_F=2.5V$)	
Auxiliary diodes (D_{a1} , D_{a2})	FEP16GTA(400V, 16A, $V_F=1.3V$)	
Main transformers (T_1 , T_2)	Core: EE7072, Turn-ratio: 0.61	
	For T_1 ,	For T_2 ,
	L_m :855.8 μ H L_k :8.46 μ H	L_m :845.2 μ H L_k :8.32 μ H
Output inductor (L_o)	200 μ H, MPP core	
Output capacitor (C_o)	47 μ F/650V	
Controller	TMS320F28027	

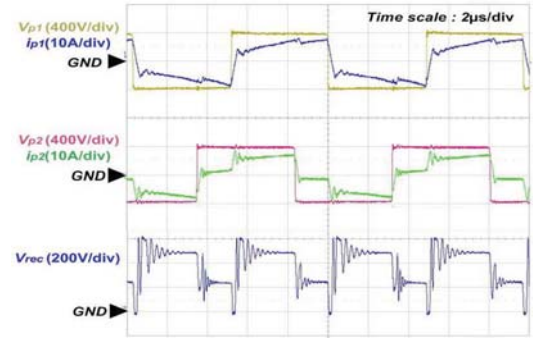


Fig. 8 Operating waveforms when the proposed converter is moving from point B to point C.

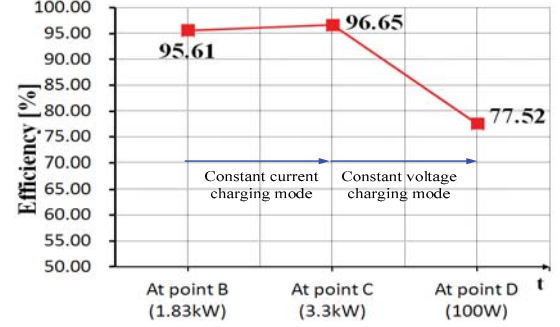


Fig. 9 Efficiency at the points B, C and D.

damaging the battery until each battery cell voltage reaches to the threshold voltage or the battery voltage reaches to the point A. This charging sequence is defined as pre-charging mode. After the pre-charging mode, the DC-DC converter is switched to constant-current mode and the battery voltage is linearly charged up to the maximum battery voltage, i.e. 450V or the point C. If the battery voltage reaches to the point C, the DC-DC converter lies finally under constant-voltage mode and the charging current gradually decreases. This mode is stopped when the charging current reaches to the pre-defined termination current as indicated in Fig. 6.

Fig. 7 shows the operations, relevant operating waveforms, and equivalent circuits of the proposed converter over the battery charging profile shown in Fig. 6. For the pre-charging mode where the processed power is small, the leading full-bridge inverter in the proposed converter is disabled and only the lagging full-bridge inverter charges the battery by adjusting the phase-shifted time of T_a in between the two legs in the lagging full-bridge inverter indicated in Fig. 7(a). That is, in this mode, the lagging full-bridge inverter operates like the conventional single PSFB converter, and the phase-shifted time of T_a decreases as the output voltage or battery voltage increases during the pre-charging mode. When the battery voltage reaches to the point A, the T_a becomes zero degree and the gate signal of Q_1 (or Q_2) is identical with of Q_3 (or Q_4). At the same time, the leading full-bridge inverter is enabled and the proposed converter starts to work by adjusting the phase-shift time of T_ϕ between the two inverters as in Fig. 2. At the point B, most of the power is transferred to the battery through the lagging full-bridge inverter as seen in Fig. 7(b) because the battery voltage is still low or the phase-shifted time of T_ϕ is small. As the charging point moves from A to C, the battery voltage or the charging power increases and T_ϕ also increases. Then, the power processed by the leading full-bridge inverter

gradually increases, and each inverter naturally handles half the total power at the maximum power point, C. These operations make it possible for the proposed converter to achieve higher efficiency over the whole battery charging process.

IV. EXPERIMENTAL RESULTS

In order to confirm the feasibility of the proposed converter, a scale-downed prototype with the power-capacity of 3.3-kW was built with the following specifications.

- $V_{IN}=400$ V, $V_O=250$ -450 V, $I_{O(max)}=7.33$ A
- Switching frequency: $f_s=100$ kHz

Table I shows the components used in the prototype converter.

Fig. 8 shows the key operating waveforms when the proposed converter is moving from B to C, which is indicated in Fig. 6. As shown in the figure, all the measured waveforms are well following the theoretical waveforms described in Figs. 2 and 7. Moreover, it is confirmed that the proposed converter has no circulating current on the primary side during battery charging. We can also see that from the rectifier output voltage waveform in Fig. 8, the power is transferred using two low-voltage rating diodes with smaller on-state voltage when the proposed converter moves from the point B to C. Due to this operation, the conduction loss generated in the rectifier stage is improved.

Fig. 9 shows the measured efficiencies. As in Fig. 9, the proposed converter has the maximum of 96.65% at a full load or the point C in Fig. 6, and the high efficiency is maintained during the battery charging. This is due to ZVS operation under all operating conditions, the design without the effect of duty loss, much reduced circulating current, and the usage of two lower-voltage rating diodes.

V. CONCLUSION

This paper proposes a soft-switching DC-DC converter for high-power battery charger with the power-capacity over 6.6-kW for EVs. The theoretical analysis of the proposed converter was presented, and its feasibility was verified through the experiment with a scale-downed 3.3-kW prototype converter. The experimental results shows that the proposed converter is suitable for the DC-DC converter applications in high-power battery charger with the power-capacity of over 6.6-kW for EVs.

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